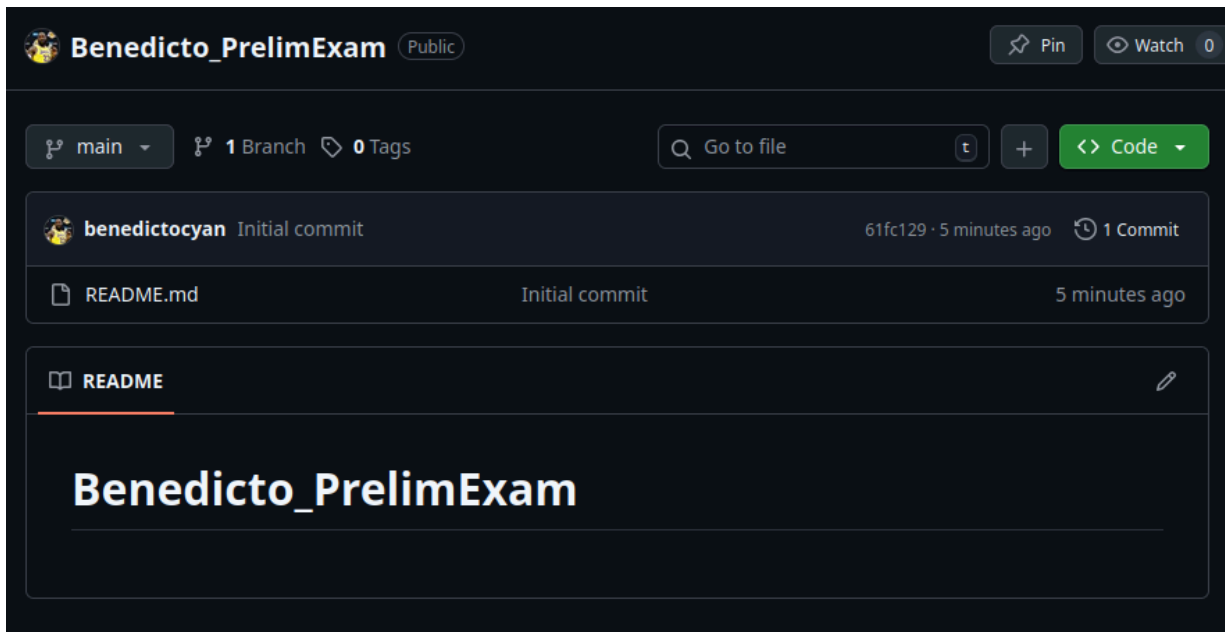


PrelimExam	
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Section: CPE31S1	Instructor: Engr. Robin Valenzuela

Create a repository in your GitHub account and label it as Surname_PrelimExam



Verification that I already cloned github in my control node.

```

benedicto@Workstation:~$ ls
Benedicto_PrelimExam  CPE212_Benedicto
cpe212-               Desktop
CPE212                Documents
CPE212_31S1           Downloads
benedicto@Workstation:~$
  
```

```
! Playbook.yml U  ≡ inventory.ini U X
≡ inventory.ini
1  [Ubuntu]
2  192.168.56.104
3  192.168.56.103
4
```

```
! Playbook.yml U  ⚙ ansible.cfg U X
⚙ ansible.cfg
1  [[defaults]]
2  inventory = inventory.ini
3  Private_key_file = /home/user/.ssh/id_ed25519
4
```

Starting the playbook with name of what will be doing in this .yml file followed by hosts where the tasks will be downloaded and then become: yes for sudo

```
---
- name: Installing essentials |
  hosts: Ubuntu #Hosts where all the tasks will be download
  become: yes
```

Here I downloaded the python and python3-pip using the ansible loop. I tried it first before adding the remaining essentials that I needed to download. That's why in the next screenshot the python3 and python3-pip is already installed.

```
benedicto@Workstation:~/Benedicto_PrelimExam$ ansible-playbook Playbook.yml -K
ok: [192.168.56.103] => (item=python3)
changed: [192.168.56.104] => (item=python3-pip)
changed: [192.168.56.103] => (item=python3-pip)

PLAY RECAP *****
192.168.56.103      : ok=2    changed=1    unreachable=0    failed=0    skipped=0
nored=0
192.168.56.104      : ok=2    changed=1    unreachable=0    failed=0    skipped=0
nored=0

benedicto@Workstation:~/Benedicto_PrelimExam$
```

The loop is essential to make repetitive tasks like downloading in my case. It made my code shorter and easier to understand.

```
1 # Playbook.yml
2 - name: Installing essentials
3
4
5
6 tasks:
7   - name: Install python3, python3-pip
8     apt:
9       name: "{{ item }}"
10      state: present
11      update_cache: yes
12    loop:
13      - python3
14      - python3-pip
15      - mariadb-server
16      - openjdk-11-jdk
17   - name:
18
19
20
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
benedicto@Workstation:~/Benedicto_PrelimExam$ ansible-playbook Pl
changed: [192.168.56.104] => (item=openjdk-11-jdk)
changed: [192.168.56.103] => (item=openjdk-11-jdk)

PLAY RECAP *****
**
192.168.56.103          : ok=2    changed=1    unreachable=0
ignored=0
192.168.56.104          : ok=2    changed=1    unreachable=0
ignored=0

benedicto@Workstation:~/Benedicto_PrelimExam$
```

Starting the mariadb and java script using service:, state started and enabled: yes so that when I enter systemctl the package is already enabled and started.

```
- name: Start
  service:
    name: mariadb
    state: started
    enabled: yes

- name: Start
  service:
    name: openjdk-11-jdk
    state: started
    enabled: yes

- name: Create table
  mysql_query:
```

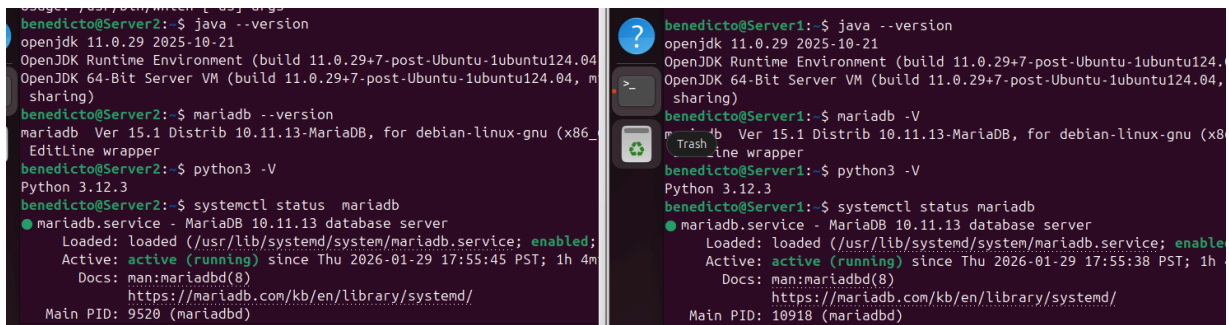
Code for creating table but I don't have proof that It worked in managed nodes

```
- name: Create table
mysql_query:
  db: benedicto_db
  sql: |
    CREATE TABLE IF NOT EXISTS users (
      id INT AUTO_INCREMENT PRIMARY KEY,
      name VARCHAR(100),
      section VAR(20)
    );
    INSERT INTO users (name, section) VALUES ('Benedicto', 'CPE31S1')
```

Create user code but doesn't work.

```
- name: Create a user
user:
  name: BenedictoJr
  shell: /bin/bash
  home: /home/BenedictoJr
```

Proof:



The image shows two terminal windows side-by-side. The left window is on 'benedicto@Server2' and the right is on 'benedicto@Server1'. Both show the same sequence of commands: 'java --version', 'mariadb --version', 'python3 -V', and 'systemctl status mariadb'. The output for 'systemctl status mariadb' shows the service is 'loaded' and 'active (running)'.

```
benedicto@Server2:~$ java --version
openjdk 11.0.29 2025-10-21
OpenJDK Runtime Environment (build 11.0.29+7-post-Ubuntu-1ubuntu124.04)
OpenJDK 64-Bit Server VM (build 11.0.29+7-post-Ubuntu-1ubuntu124.04, m
sharing)
benedicto@Server2:~$ mariadb --version
mariadb Ver 15.1 Distrib 10.11.13-MariaDB, for debian-linux-gnu (x86_
Editline wrapper
benedicto@Server2:~$ python3 -V
Python 3.12.3
benedicto@Server2:~$ systemctl status mariadb
● mariadb.service - MariaDB 10.11.13 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled;
   Active: active (running) since Thu 2026-01-29 17:55:45 PST; 1h 4m
   Docs: man:mariadb(8)
         https://mariadb.com/kb/en/library/systemd/
   Main PID: 9520 (mariadb)

benedicto@Server1:~$ java --version
openjdk 11.0.29 2025-10-21
OpenJDK Runtime Environment (build 11.0.29+7-post-Ubuntu-1ubuntu124.
OpenJDK 64-Bit Server VM (build 11.0.29+7-post-Ubuntu-1ubuntu124.04,
sharing)
benedicto@Server1:~$ mariadb -V
mariadb Ver 15.1 Distrib 10.11.13-MariaDB, for debian-linux-gnu (x86
Editline wrapper
benedicto@Server1:~$ python3 -V
Python 3.12.3
benedicto@Server1:~$ systemctl status mariadb
● mariadb.service - MariaDB 10.11.13 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enable
   Active: active (running) since Thu 2026-01-29 17:55:38 PST; 1h
   Docs: man:mariadb(8)
         https://mariadb.com/kb/en/library/systemd/
   Main PID: 10918 (mariadb)
```

Personalize git before commit and push

```
benedicto@Workstation:~$ git config --global user.name "Benedicto"
benedicto@Workstation:~$ git config --global user.email "qrabenedicto@tip.edu.ph"
benedicto@Workstation:~$ cat ~/.gitconfig
[user]
    name = Benedicto
    email = qrabenedicto@tip.edu.ph
benedicto@Workstation:~$
```

Git commit, git push

```
• benedicto@Workstation:~/Benedicto_PrelimExam$ git add *
• benedicto@Workstation:~/Benedicto_PrelimExam$ git commit -m "PrelimExam"
[main aaf12cf] PrelimExam
 4 files changed, 76 insertions(+)
 create mode 100644 Playbook.yml
 create mode 100644 ansible.cfg
 create mode 100644 inventory.ini
 create mode 100644 prac.yml
• benedicto@Workstation:~/Benedicto_PrelimExam$ git push
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 4 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 1.12 KiB | 1.12 MiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0
```

Github repo: https://github.com/benedictocyan/Benedicto_PrelimExam

Conclusion: In conclusion, knowing the structure of the playbook is very helpful in doing some of the activities or tasks here but some are not that easy to do. I'm having a hard time understanding what to do or how to start on doing them. Overall this exam made me want to study more about automating servers specially ansible playbooks.